

Packet loss can be replicated:

```
# tc qdisc change dev eth1 root netem loss 0.1%
```

Packet duplication/corruption can also be configured:

```
# tc qdisc change dev eth1 root netem duplicate/corrupt 1%
```

Setting up the throttling

In the current environment (as seen in Figure 1), the Linux server plays the role of a router and a NetEM bandwidth throttling device. All the traffic goes through this server. Our requirement was to throttle the bandwidth by adding delay/packet loss/jitter etc, and also to throttle incoming traffic on TCP port 7001 on *eth1*, down to 512 kbit/s.

To set up the throttling for incoming traffic, I ran the following commands:

```
# tc qdisc add dev eth1 root handle \
1: cbq wqpt 1000 bandwidth 10Mbit
# tc class add dev eth1 parent 1: classid 1:1 \
cbq rate 512kbit allot 1500 prio 3 bounded isolated
# tc filter add dev eth1 parent 1: protocol ip u32 \
match ip protocol 6 0xffff match ip dport 7001 0xffff flowid 1:1
```

The first command created a new root queuing discipline of type CBQ (class-based queuing, used to both shape and

prioritise) on interface *eth1*, bandwidth set to 10 MBps, and the average size of the packet (*avpkt*) set to 1000.

The second command created a child class under the root *qdisc*, 1:, and named it 1:1, allocating 1500 bytes off the queue, and limiting the rate to 512 KBps. The *prio* 5 defines the priority of the queue. The smaller the number, the higher the priority when you have multiple queues with different priorities. Here I had only a single queue, so that didn't matter. The *bounded* and *isolated* options are used to make sure the class will not borrow bandwidth from any unused/inactive sibling queues.

The last command is the filter, which will queue all packets going to TCP port 7001 [*dport* is destination port] to *qdisc* "1:1" which is the throttle queue. Here *u32* is the filter type.

NetEM is vast, and we can't cover everything in this article. For more information, read the man page for *tc* (*man tc*) and see <http://lartc.org/howto/lartc.qdisc.classful.html>. **END** 

By: Pradyumna Kumar Dash

The author is senior consultant at Capgemini Consulting India Pvt Ltd, Kolkata, on the Infrastructure Transformation Services team. An open source fanatic, he loves to cook up both food and ideas. He can be contacted at: neomatrixgem@gmail.com

www.eleb2b.com

Electronic Components Products Machinery Will now be sold here

Presenting Electronics Bazaar,
India's First B2B Sourcing Guide
For Electronics Components,
Products and Machinery.

Unit Price: **Rs 50**
Subscription Rate: **Rs 350 (12 Issues)**

EFY Enterprises Pvt Ltd, D-87/1, Okhla Industrial Area,
Phase-4, New Delhi 110 020, Ph: 011-26810601-03; Fax: 26817563;
Email: info@efyindia.com, Website: efyindia.com



Only a specialist can make you a Winner



redhat
CERTIFIED
TRAINING PARTNER

Best Certified Red Hat Training Partner, South
India for 3 times in a row

RHCE of the year - 09, bagged by a
COSS student

COSS produces more than 500
RHCEs in a year

11 RHCSs certified from COSS,
the highest in India

Consistent 97% passing rate in
RHCE examination

Undisputed leader in South India
for Open Source Training



RHCE · RHCS · RHCA
SOLARIS · VERITAS · JBOSS



Complete Open Source Solutions

512, Aditya Trade Centre, Ameerpet, Hyderabad - 500038
Tel: +91-40-66773365, Fax: +91-40-66463365
Email: contact@cosindia.net
URL: www.cossindia.net